

Quiz 3 Answers

1. B) Clustering groups data points based on their similarity, creating cohesive groups. Explanation: Clustering involves grouping data points so that points within the same group are more similar to each other than to points in other groups.

2. A) Minimizing the variance within each cluster to ensure tighter groupings. Explanation: The k-means algorithm aims to make points within each cluster as close as possible to each other, minimizing the within-cluster variance.

3. C) Dependence on initial cluster centers and data assumptions. Explanation: K-means is sensitive to initial cluster placements and assumes clusters are spherical, affecting the results.

4. A) Segmenting images based on color or texture. Explanation: K-means is commonly used for image segmentation tasks, grouping similar pixels based on features like color.

5. A) GMM models data as a mix of multiple Gaussian distributions. Explanation: Gaussian Mixture Models represent data as a combination of several Gaussian distributions, capturing complex data patterns.

6. A) Detecting anomalies in data patterns. Explanation: GMMs can model normal data distribution and identify anomalies as data points that don't fit the learned distribution.

7. B) By identifying data points that don't align with the expected distribution. Explanation: Anomaly detection using GMM involves identifying points that deviate significantly from the learned distribution.

8. C) To improve the accuracy and relevance of clustering results. Explanation: Choosing an appropriate number of clusters helps in accurately representing the data structure without overfitting.

9. A) The Elbow Method, which visualizes variance reduction. Explanation: The Elbow Method helps determine the optimal number of clusters by identifying a point where variance reduction diminishes.

10. B) Silhouette analysis, assessing cohesion and distance between clusters. Explanation: Silhouette analysis evaluates the compactness and separation of clusters to help find the optimal number of clusters.